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Machine learning-integrated random shape generation for brand-styled mouse design

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ABSTRACT
This investigation aims to develop a novel design methodology that enables industrial designers to incorporate machine learning into the process of generating random shapes, thereby facilitating the creation of products with a consistent brand style. The primary distinctions between this approach and traditional design procedures include the utilisation of a substantial number of 2D and 3D samples generated through random shape generation. This study employs a computer mouse as a case example. Following four stages – production, measurement, evaluation, and analysis – the optimal design of the mouse is ultimately determined. The experimental design involves the random generation of 10,000 samples, divided into 50 groups. Subsequently, these samples are assessed using a deep learning-based brand classification system, which classifies them into specific brand styles. Next, a questionnaire survey has been conducted to compare mice created by random shape generation technology with those designed by designers, examining the differences in appearance and usability between these two design methodologies. The results show that the mouse generated via random shape generation technology surpasses the designer's design in both appearance and usability, with statistically significant differences. This method is worth promoting as part of the company's brand strategy and for the development of future new products.

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KEYWORDS
Random shape generation; machine learning; brand styling classification system; deep learning; t-SNE

1. Introduction

Recently, AI-based generative design has become increasingly widespread. However, its application in the field of industrial design still faces four significant challenges, including concept generation, prompt development, from 2D concept image creation to 3D modelling, and ultimately producing products with a specific brand style. The first challenge lies in converting 2D images into 3D models. If reverse engineering techniques are used at this stage, the resulting 3D models converted from 2D images may suffer from poor quality. We aim to explore new approaches to address this issue. The second challenge is the development of innovative concept designs. As design automation technologies continue to advance, they can generate a wide variety of product concepts. However, the quality of

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these generated designs still needs to be evaluated. The third challenge involves selecting optimal design solutions. When generative design is used to create a large number of alternatives with a specific brand style, a systematic method is needed to evaluate and select the optimised design. The final challenge is determining whether a product truly embodies a specific brand style. This involves using machine learning and deep learning to verify whether the optimised design aligns with the intended brand identity. In response to these challenges, we review relevant theories spanning a century, from formalism and design automation to machine learning. Our goal is to explore an innovative product design methodology that integrates AI into random shape generation to support brand-style product design, selection, and decision-making. The research objectives are as follows:

- To address the issue of converting 2D images into 3D models in random shape generative design
- To use random shape generation techniques to assist in generating diverse product shapes and concepts
- To apply AI techniques to filter and select from the numerous generated samples
- To verify whether the optimised design conforms to a specific brand style

The remainder of this paper is organised as follows. Section 2 presents a comprehensive literature review, covering topics such as formalism, product design, brand style, shape grammar, shape design automation, random shape generation, and the application of machine learning in product design. Section 3 introduces the research methodology and steps. The first half of Section 4 describes the case studies, measurements, and analytical evaluations, including the generation of samples using random shape generation, area measurements of samples, 2D/3D CAD surface conversions, and the resulting mouse appearance designs. The latter half presents results from predicting new product design styles and selecting the optimised design using a brand style classification system. Section 5 presents practical design cases that utilise random shape generation, explaining the selection of samples based on a brand-style recommendation system and predicting the style of 3D models using t-SNE. This section also explores brand style differences and consistencies, as well as the application of AI in product design through the generation of random shapes. The last section of this section outlines a survey on mouse models designed using two methods: random shape generation technology and design by designers. Section 6 summarises our findings and provides suggestions for future research directions.

2. Literature review

2.1. Formalism

Formalism, a critical methodology that emerged in the late 19th and early 20th centuries, asserts that artistic and design value derives primarily from form and structure rather than external context (Bois et al. 2004; Clarke 2010; Maghsoudlou and Nouran 2020; Pinotti 2012). Grounded in Kant's theory of beauty and the autonomy of art, this perspective was reinforced by semiotics, which emphasised symbols independent of meaning.

Closely linked to modernism and the International Style (Levine 1986), formalism promoted 'art for art's sake' and strongly influenced abstract movements such as Impressionism, Fauvism, Expressionism, and Cubism (Flanagan 2002; Kandinsky 1977; Kramer 1973). In product design, Bauhaus exemplified formalist ideals, where Riegl's concept of Kunstwollen framed each era as seeking unique forms, and Schapiro emphasised style as enduring structure (Parker 1921; Pinotti 2012; Riegl and Kemp 1995). Contemporary analysis, such as Starck's armchair (Wang 2022), shows the continuing influence of this orientation.

By equating style with stable visual features, formalism laid the foundation for computational approaches including shape grammar and pattern languages (Stiny 1980, 2006; Wortmann 2013). These frameworks formalised design into sets of rules and operations, enabling systematic generation of stylistically coherent outcomes (Jowers, Earl, and Stiny 2019). Consequently, formalism's legacy extends naturally to artificial intelligence and machine learning, where algorithms generate and evaluate forms through structured rules, bridging traditional aesthetics with contemporary generative design.

2.2. The relationship between brand style and product design, and its evaluation

Design has historically bridged art and engineering (Keitsch and Prestholt 2015). Industrial design discussions began in Britain in the late nineteenth century, followed by early twentieth-century Germany, which became the birthplace of modern industrial design (Bürdek 2002; De Fusco 2005). Contemporary perspectives emphasise the relationship between design and mechanisation (Figueiredo, Correia, and Secca Ruivo 2016). Before World War II, only a few hundred industrial designers worked in the United States, primarily focusing on consumer product styling to support marketing. Consequently, design education emphasised four domains: art, engineering, economics, and humanities (Lippincott 1945). From a modernist standpoint, designers were seen as translators of ideas into coherent visual communication (Bierut, Drenttel, and Heller 2006). As such, product design emerged as a multidisciplinary practice combining art, engineering, market research, and data analysis (de Vere, Melles, and Kapoor 2009).

With the shift toward knowledge-based processes, design theory and AI problem-solving informed new models such as decomposition, case-based reasoning, and transformation (Maher 1990). Computational tools supported exploration and solution generation (Kurtoglu, Swantner, and Campbell 2008), while biomimicry enriched creativity by offering biological inspiration (Jahed 2025). Product creativity has often been evaluated using the Creative Product Semantic Scale (CPSS), emphasising novelty, resolution, and style (Brown 2015). Multidisciplinary Design Optimization (MDO) further expanded design exploration and efficiency, particularly in aerospace applications (La Rocca and Van Tooren 2007, 2010). Together, these approaches highlight the ongoing integration of diverse tools to enhance creativity, quality, and consistency with brand values.

Three-dimensional modelling is central to contemporary product development. Realistic shapes improve simulation accuracy (Gruyters et al. 2018). Fischer, Holder, and Maier (2022) argued that 3D modelling plays a critical role in shaping both product appearance and brand identity by enabling the deliberate design of products that evoke specific perceptions and emotional responses. Conditional generative methods, such as point-cloud – based approaches, balance shared product DNA with novel suggestions (Krahe et al. 2020).

Thus, CAD and 3D modelling play a critical role in shaping both product appearance and brand identity.

From a theoretical perspective, style is often regarded as enduring yet inherently ambiguous. The European tradition distinguished between transient fashion and lasting style (Gronow 1993). Contemporary research emphasises adaptability, noting that consumers assign brand identity based on perceived similarity (Person and Snelders 2010). Fischer, Holder, and Maier (2020) classified similarity into four brand levels – historical connections, product portfolio, product family, and competitor brands – and four substructures: layout, shape, colour, and graphics. These frameworks underscore the importance of style consistency for product design and brand building.

2.3. Shape grammar, shape design automation, and shape random generation

Shape grammar provides a formal framework for generating shapes, where dynamic transformations of structures enable ambiguity and emergence, making it a strong foundation for computer-aided design exploration (Emdanat and Vakalo 1996; Jowers, Earl, and Stiny 2019; Stiny 2006). Unlike equation-based methods, shape grammar treats design as 'computing with shapes', defined through labelled and parameterised rules (Stiny 1980). This approach formalises design styles into visual rule libraries (Wortmann 2013), allowing designers to generate and analyze forms in systematic ways.

Cagan (2001) identified four reasons for the significance of shape grammars: (a) modelling and searching large design spaces, (b) supporting 'design-to-order' by ensuring consistent styles, (c) enabling representation and analysis, and (d) providing educational value. Evaluation mechanisms have also been proposed, such as Cui and Tang's (2013) framework assessing subjectivity, structure, elements, and complexity. Initially applied in architecture, shape grammar supported projects like Palladian floor plans (Stiny and Mitchell 1978) and later extended to product design, including coffee makers (Agarwal and Cagan 1998), multifunctional chairs (García and Menezes Leitão 2018), vases (Mata, Ahmed-Kristensen, and Shea 2018), and crossover vehicles (Orsborn et al. 2006). Brand-specific cases include Harley-Davidson motorcycles (Pugliese and Cagan 2002) and Coca-Cola bottles (Ang et al. 2006).

Recent studies integrate AI with shape grammar. For example, Zhang and Willis (2024) proposed an approach – fusing a formal generative model for 3D shapes with deep learning (DL) – to improve the understanding and generation of 3D object geometry by combining human-defined shape grammar rules with data-driven AI methods. Berčič (2024) implemented shape grammar within parametric tools such as Rhinoceros 3D and Grasshopper, highlighting how AI and machine learning-based decision systems can filter and select optimal design alternatives generated by shape grammars. Çelik (2024) applied shape grammar to the generation and evaluation of architectural plan diagrams using text-to-image AI algorithms, while Alcaide-Marzal, Diego-Mas, and Acosta-Zazueta (2020) proposed grammar-based sketch transformation with parametric modelling. Other efforts include topology optimisation for wheel design (Shin et al. 2021), reinforcement learning for maximising design diversity (Jang, Yoo, and Kang 2022), and AI-powered product identity design using shape grammar and affective engineering (Wang et al. 2024). Collectively, these systems demonstrate shorter design cycles, improved quality, and higher efficiency compared to rule-based heuristics.

Overall, automated shape design and random shape generation share technical similarities but differ in purpose. Automation emphasises optimisation under constraints, often serving engineering goals, while random generation emphasises creative exploration. For early product concept development, random shape generation provides greater potential to inspire diverse forms while maintaining stylistic coherence.

2.4. Applications of machine learning and deep learning in product design

Traditional product design relies heavily on designer experience, which introduces subjectivity and risks overlooking key factors. At the same time, the design process generates large volumes of data that can inform more systematic decision-making. To address this, researchers have applied machine learning and deep learning to enhance decision support. For instance, Lai et al. (2021) proposed a deep neural network – based method for complex product design decisions, while Huang, Chang, and Arinez (2020) introduced a hybrid approach that predicts product completion time by integrating machine learning with system models. In emotional design, Chan et al. (2020) combined big data and machine learning to capture user preferences and inform product aesthetics.

As AI technologies matured, integration with CAD and CAE systems became more prominent. Yoo et al. (2021) developed a deep learning framework that can automatically generate 3D CAD models, predict CAE outcomes in real time, and verify results. Burnap et al. (2016) modelled product design space distributions using extensive image and attribute data, while Hunde and Woldeyohannes (2022) reviewed AI-enhanced CAD applications and future prospects. Krahe et al. (2020) further demonstrated conditional generative methods for balancing shared product DNA with novel designs.

Beyond CAD, deep learning has been applied to generative design and recommender systems. Generative design aims not for a single optimum but for multiple feasible alternatives (Kallioras and Lagaros 2020). In recommender contexts, Turkut et al. (2020) developed a textile pattern system, Jo et al. (2020) proposed a fashion retrieval tool that pairs tops and bottoms, and Masakazu and Tomokoi (2020) introduced a sensibility-based system that aligns recommendations with aesthetic preferences. These efforts demonstrate how machine learning can expand design exploration and personalisation.

Although many applications remain experimental, some have demonstrated practical impact. In a curtain airbag case study, Arjomandi Rad, Cenanovic, and Salomonsson (2023) generated 60,000 CAD samples through design automation and used deep learning to filter and evaluate them. This approach reduced verification time and enabled predictive modelling in early design phases. Collectively, such studies highlight the potential of data-driven, AI-supported approaches to foster innovation, improve efficiency, and support more informed product development.

2.5. Summary

There exists a close relationship between AI-generated imagery, brand style, and product design, which directly shapes how manufacturers approach form development. First, AI-generated imagery can ensure brand style consistency (Wang and Chen 2022), enhance efficiency, and improve the adaptability of aesthetics. As noted in the discussion of formalism (Section 2.1), AI models can be trained to replicate brand-specific elements such as colour schemes, styles, and themes, ensuring alignment with established identities. For instance, convolutional neural networks have been used to classify movie posters by genre, capturing visual appearance and object features (Chu and Guo 2017).

Second, AI-generated imagery expands prototyping, customisation, and design exploration. Designers can employ AI tools to create rapid product visualisations, support customisation, and stimulate novel ideas that may not emerge from human-only processes. For example, Ishihara, Kuo, and Ishihara (2023) showed how AI-supported imagery enhances affective engineering and inspires innovative product forms.

Its relationship with formalism can be described in four ways:

(A) Formalism as a basis: AI models can reproduce formal principles, producing forms that emphasise balance and composition.
(B) Expanding aesthetics: AI enables exploration of new combinations beyond traditional formal boundaries.
(C) Challenging interpretation: AI-generated forms raise questions about the roles of designer intentions and user perception.
(D) Hybrid approaches: AI can integrate formal principles with other stylistic methods to create mixed design outputs.

In summary, AI-generated imagery represents a methodological bridge between formalist theory and computational design practice. It enables designers to imitate, expand, or challenge traditional aesthetics while providing new tools to strengthen brand resonance, improve efficiency, and expand visual exploration.

3. Research method

This study integrates machine learning into random shape generation techniques to produce brand-style-oriented products, using computer mice as a case study. The process involves four stages: generation, measurement, evaluation, and analysis. The core idea is to employ machine learning methods to filter the random shape generation mouse shapes, selecting those that optimally align with the target brand style. Additionally, a questionnaire survey was conducted to compare the two design approaches: computer mice designed through random shape generation with those designed by human designers, evaluating differences in appearance and ergonomic usability.

3.1. Design process

We propose a design process that begins with random shape generation and ends with final product modelling. The distinguishing feature of this approach is its integration of generative and classification systems for designing brand-style products. The process is divided into three main parts: (A) Random Shape Generation Technique, (B) Brand Style and Product Design, (C) Deep Learning-Based Brand Style Evaluation, as shown in Figure 1.

(A) Random Shape Generation Technique

This phase outlines the steps involved in generating and selecting shapes using random shape generation techniques. The literature review includes formalism (Section 2.1),

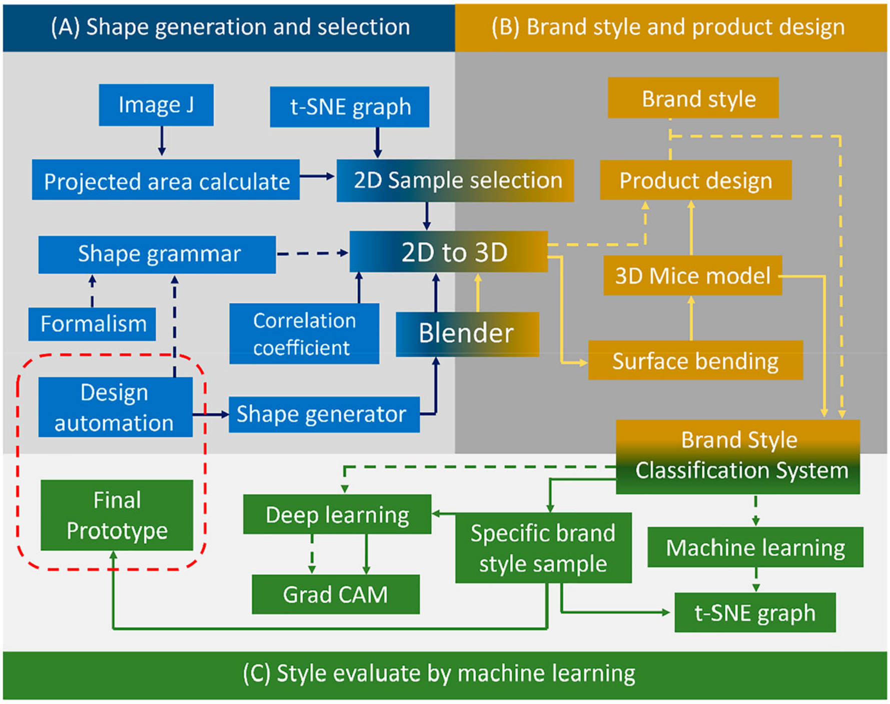

Figure 1. Generative and classification design process from abstract style to concrete modelling process.

shape grammar and random shape generation techniques (Section 2.3). In Figure 1, the dashed lines indicate relationships established from the literature, while solid lines represent the experimental design process of this study. Starting from random shape generation techniques (Section 2.3), shape generation (Section 3.2). This is then extended to the association and conversion between 2D and 3D (Section 4.1), which is based on the statistical foundations discussed in correlation and regression analysis (Section 3.3). It also involves calculating 2D projected areas (Section 4.2), sample selection, and t-SNE visualisation (Section 4.3).

(B) Brand Style and Product Design

This stage presents the design process that moves from random shape generation to 2D-to-3D conversion, and from 3D modelling to final product design. The literature review includes Section 2.2 (brand style and product design). The experimental components comprise 2D-to-3D transformation (Section 4.1), surface deformation, and 3D modelling for mouse design (Section 4.4), which prepares the designs for brand style classification (Section 4.5).

(C) Deep Learning-Based Brand Style Evaluation

In this phase, we apply the previously developed 'brand style classification system'. The literature review includes machine learning and deep learning (Section 2.4). The study involves applying previous findings (Wang and Chen 2022) through t-SNE, a non-linear machine learning dimensionality reduction method, to classify newly designed mice (Section 4.3). Experimental procedures include brand style classification system (Section 4.5), samples with specific brand styles, and the use of Grad-CAM heatmaps to visualise and filter the generated designs (Section 4.6), ultimately resulting in the final model (Section 5.1).

This random shape generation design process for classification of product brand styles includes a literature review and actual design experimentation. Dashed lines represent the causal paths informed by literature and conceptual reasoning, while solid lines indicate the sequential steps of the design experiment. The study process is a knowledge-based design. Given that random shape generation produces a large number of samples, the central research question of this study is how to effectively select the optimal designs that conform to specific brand styles from a large dataset.

The design experiment consists of the following ten steps, as illustrated in Figure 2. The first step is to randomly sample from a large data set based on 200 model samples. The second step is to generate samples. In this study, we are using the Shape Generator add-on in Blender to generate 2D and 3D shapes. In the third step, we generate various shapes based on the numbers we provided from 6801 to 7000. For example, when we give the number 6907 to the Shape Generator add-on of Blender. It will generate a 2D image and a 3D model of the number 6907 according to the previously specified parameters. In simple terms, this stage finds the 2D/3D correlation of the 200 samples. The fourth step is to save the 2D image, the 3D model, and the corresponding 2D image of the 3D model, before linking all three objects to the number. This stage lists the sample numbers, and the tool is Excel. The fifth step involves sample measurement, using the image analysis software ImageJ, developed by the National Institutes of Health (NIH), to calculate the projected area of each of the 200 samples. This process involves filtering out the smaller ones and selecting the sample with the larger area.

The sixth step is to use t-SNE to analyze the 2D samples of the generated data group 35; the distribution of the 200 samples. In this stage, we use t-SNE to analyze all randomly generated 2D samples. The seventh step converts the generated 2D/3D CAD into a surface and creates the shapes of four computer mice. The eighth step utilises the brand style classification system to evaluate the brand style and generates a heat map prediction using deep learning for the four samples. The ninth step is to select one sample and use t-SNE to evaluate the style of the generated 3D sample, which can show the distribution of this sample under the brand style classification. The tenth step is to select the sample and transform it into a prototype that can be 3d printed.

3.2. Random shape generation

The process began by generating 10,000 samples, from which 200 were randomly selected as experiment samples. This approach prevents issues related to the excessive size of the sample set, such as difficulty in visually presenting all samples. The term ‘samples’ here refers to roughly formed but distinctive shapes generated through random shape generation techniques, rather than the traditional process of mouse shape designs. Thus, both 2D images and 3D models still require post-processing via computer-aided design tools to develop into 3D surfaces and to add necessary mechanical design to become final mouse

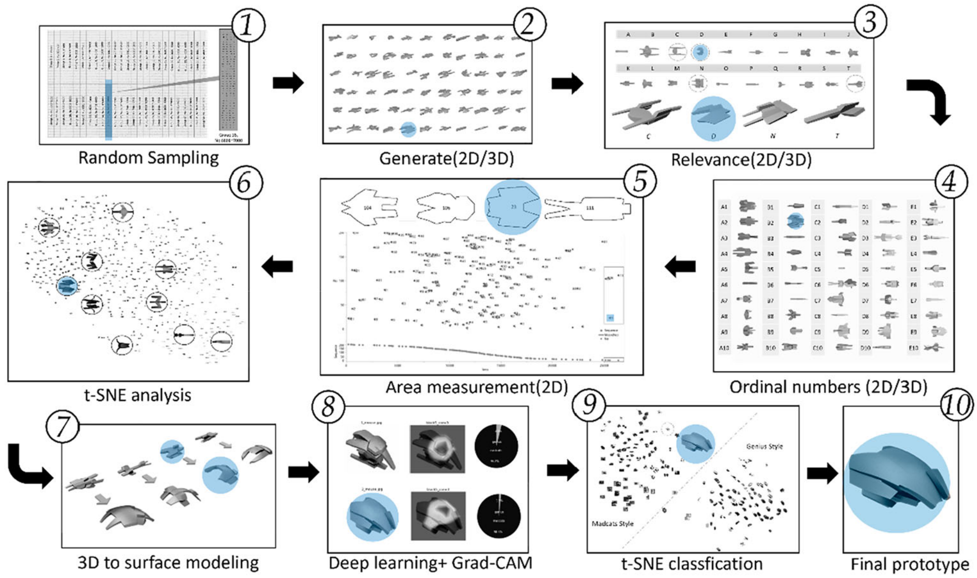

Figure 2. Steps from random generation, selection, and design of a prototype.

prototypes.

These types of samples serve as conceptual development and creative explorations during the early stages of mouse design. This method can help designers shorten the ideation and shape development process. This thereby enhances the practical value of random shape generation techniques. However, when an excessive number of shapes are generated, random sampling is used to avoid displaying overly small dots on the visual output, making individual samples more distinguishable.

Next, we used the software Blender, along with its add-on Shape Generator, to generate both 2D images and 3D models. The 10,000 samples were divided into 50 groups of 200 samples each, and we randomly selected one group for further experiment. The study applies a random number generator that generates a number from 1 to 50, then selects one of those numbers. The Group 35 was chosen in this study (see Figure 3). At this stage, both 2D and 3D models were generated using the Shape Generator add-on. The 3D models are essential for the later stages of product design, while 2D images are more suitable for identification, management, and area measurement. In other words, generating 3D models only would hinder identification and organisation, whereas generating 2D images only would prevent the evaluation of brand style and the development of final products. Thus, we need to connect 2D images and 3D models after generating random shapes.

The shape generation process was based on input parameters, including extrusion amount, extrusion length, and taper, which influence the appearance of the shapes. To simplify and reduce the computational burden during shape generation, parameters such as bevel and subdivision, which affect detail refinement, were intentionally unset. To obtain the overall 2D image of all samples, a top-view projection was rendered in Blender using the Shape Generator add-on. Additionally, individual 2D and 3D images were generated sequentially, resulting in 200 images for each (2D and 3D), as well as a combined comprehensive 2D image, as shown in Figure 4.

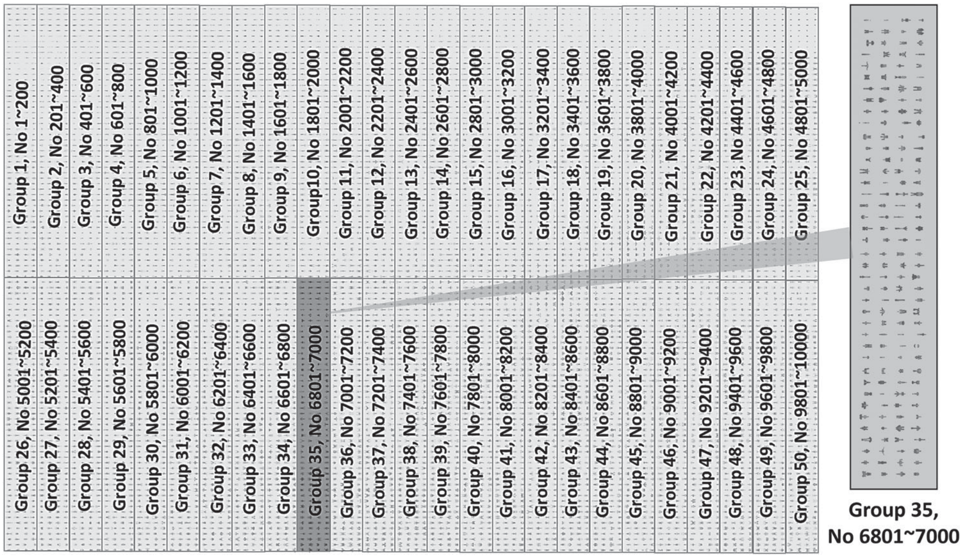

Figure 3. The 10,000 samples are divided into 50 groups, each with 200 samples. One group was randomly selected for the experiment.

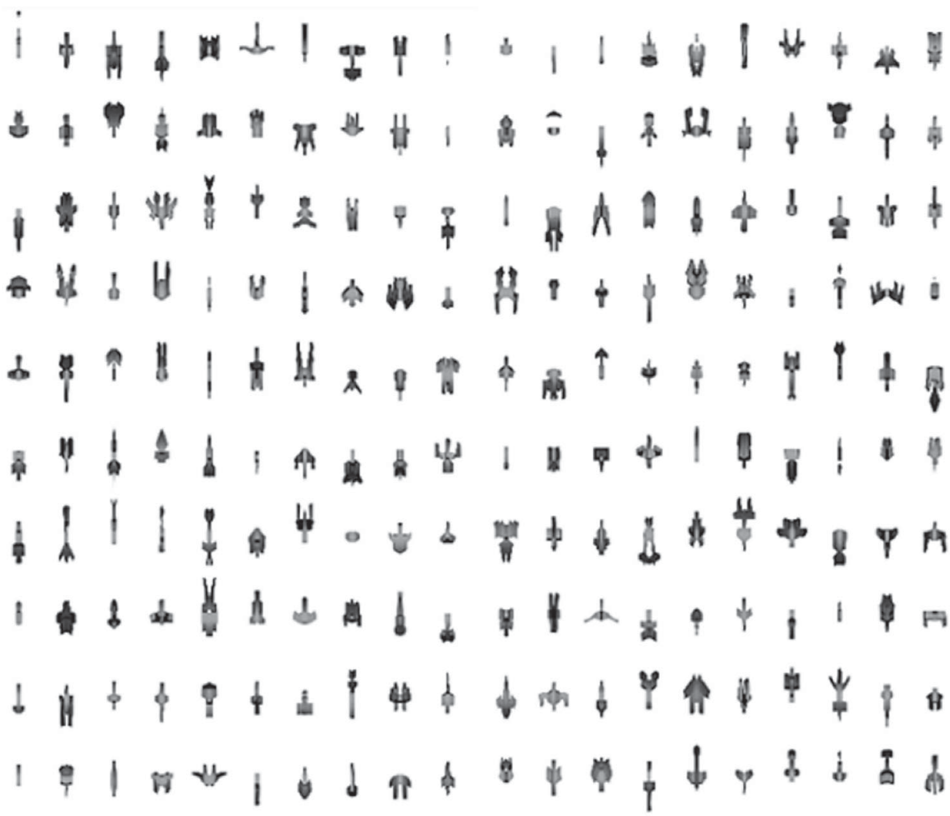

Figure 4. Randomly generated shapes (200 2D images overview).

Regarding setting up the Shape Generator add-on in Blender, we first need to install it. After installation, the Shape Generator appears in Blender, and it is also visible in the side panel. Next, we start adjusting some parameters and import the geometry node with the pre-constructed matrix. The parameters are then set to range from 6801 to 7000 (taking Group 35 as an example). This approach will generate 200 shapes randomly by the Shape Generator. The following step is to convert the 3D models into top-view and perspective images for saving. We can use the built-in Iterator function of the Shape Generator to generate sequential images.

3.3. Correlation coefficient and regression analysis

After shape generation, we analyzed the relationship between 2D profiles and the 3D surface of every model. Since the final product – a 3D model of a computer mouse with a specific brand style – is selected from 2D images generated via random shape generation, it is crucial to identify the correlation between 2D images and 3D shapes. This stage ensures that filtering by 2D projected area effectively contributes to the decision-making process for the final 3D model.

Pearson correlation and simple linear regression were selected because they are well-established, interpretable methods for continuous variables expected to have a linear relationship. Correlation quantified the strength of association between 2D and 3D areas, while regression provided a predictive equation for practical filtering of design alternatives. Other methods, such as nonparametric correlations, higher-order regressions, or machine learning models, were not employed in this study due to the small sample size and the need for a transparent, easily interpretable model.

Pearson correlation coefficient quantifies the strength and direction of the linear relationship between two variables. The coefficient ranges between −1 and 1: '1' indicates a perfect positive linear correlation, '−1' indicates a perfect negative linear correlation, and '0' indicates no linear correlation.

The Pearson correlation coefficient is commonly denoted as r, and its formula for calculating the linear relationship between two variables X and Y is as follows:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2}} \tag{1}$$

- r: Pearson correlation coefficient. It measures the strength and direction of the linear relationship between variables x and y.
- x_i: the value of variable x for the i-th sample.
- y_i: the value of variable y for the i-th sample.
- $\bar{x}$: the mean (average) of all x values.
- $\bar{y}$: the mean (average) of all y values.

The second method is regression analysis, which is used to model the relationship between a dependent variable (usually denoted as Y) and one or more independent variables (usually denoted as X). The goal of linear regression is to find the best-fitting straight line – also known as the regression line – that represents the relationship between variables.

Table 1. 2D profile and 3D surface area dimensions of model.

	Sample Plane Number[a]									
	1	2	3	4	5	6	7	8	9	10
3D surface area[b]	4634	1841	4975	3529	2594	1093	4336	1865	5785	783
2D plane area[b]	1472	873	1541	1431	1714	546	1511	915	2250	313
	Sample Plane Number[a]									
	11	12	13	14	15	16	17	18	19	20
3D surface area[b]	1118	1253	534	4554	1217	2351	1078	1676	1451	784
2D plane area[b]	652	587	261	1263	421	747	399	689	647	316

[a]Numbers 1–20 indicate a total of twenty planes.
[b]The unit of the area of the drawing is mm^2

The equation for linear regression is generally expressed as:

$$Y = \beta 0 + b1 * X + \varepsilon \tag{2}$$

Where:

- Y is the dependent variable (the variable we want to predict or explain)
- X is the independent variable (the predictor of Y)
- $\beta 0$ is the intercept, representing the value of Y when $X = 0$
- $\beta 1$ is the slope of the regression line, representing the change in Y for a one-unit change in X
- ε is the error term, which accounts for the variability in Y not explained by the regression line

This study segmented each sample into several distinct planar surfaces to conduct the analysis. We individually calculated the area of both the 2D plans and the corresponding 3D surfaces using Pro-E Creo software. The area data for these samples are presented in Table 1.

The study then used SAS JMP statistical software to calculate the correlation coefficient and conduct linear regression analysis. The results showed the correlation is as follows: Examining the relationship between the 3D surface area and the 2D plane area, the Pearson correlation coefficient was found to be 0.92, indicating a strong positive correlation between the two variables. The associated p-value was 1.135e-08, which is extremely small and provides strong evidence to reject the null hypothesis of no correlation. This result suggests that we can be 95% confident that a true correlation exists. Overall, the data indicate a statistically significant positive relationship between the 3D surface area (X) and the 2D plane area (Y). The result of the linear regression analysis is shown in Figure 5, and the regression equation is as follows:

$$Y = 0.314 * X + 183.14 \tag{3}$$

Regarding the mean probability of brand style, we calculated the brand style probabilities of the optimised samples based on predictions from the deep learning model. The mean value, Mean (Z), is derived from the average of these predicted probabilities. If there are N

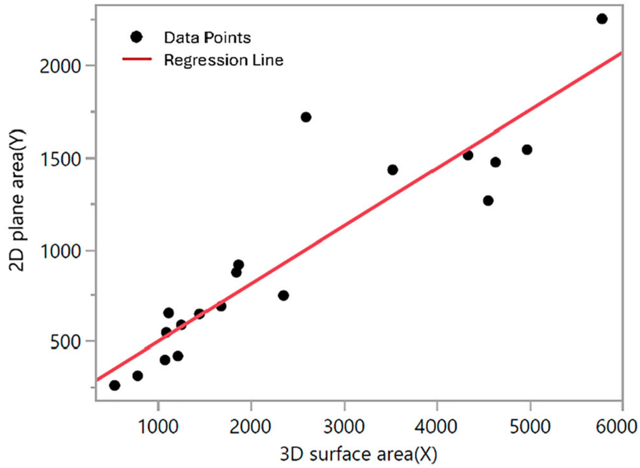

Figure 5. Result of the linear regression analysis.

optimised samples (e.g. N = 5), the formula is expressed as:

$$\text{Mean (Z)} = (Z_1 + Z_2 + Z_3 + \ldots + Z_N) / N \tag{4}$$

The correlation and regression analyses confirm that filtering based on 2D plane area is indeed helpful in determining the final 3D surface shape. Therefore, the next section will begin with the procedure for filtering samples based on their 2D plane area.

4. Case study on measuring and analyzing samples

4.1. 2d/3D correlation analysis and sample numbering

The study associated 2D and 3D images. Since we had comprehensive 2D images and corresponding IDs, we used these as references to link each of the 200 individual 2D and 3D images. As shown in Figure 6, the 2D images labelled C, D, N, and T correspond to their respective 3D images, establishing a correlation between them. A management system is established for this randomly generated shape database, which enables the establishment of relationships between 2D images, 3D images, and 3D models.

At the same time, the study linked the 2D/3D correlations for 200 of them and assigned identification numbers to the generated samples. We used Blender software along with the Shape Generator add-on and were able to generate a large number of distinct shape samples. However, due to the vast quantity, a management system was necessary. This study selected 200 samples for experimental purposes, which is easier to manage and analyze compared to 10,000 samples. These 2D and 3D images were numbered in sequence, as shown in Figure 7, where we selected 100 2D and 3D images from the entire dataset for illustration.

4.2. Sample measurement

The study utilised ImageJ software to calculate the projected 2D area of each of 200 samples. From these, we selected those with relatively larger projected areas, as smaller shapes

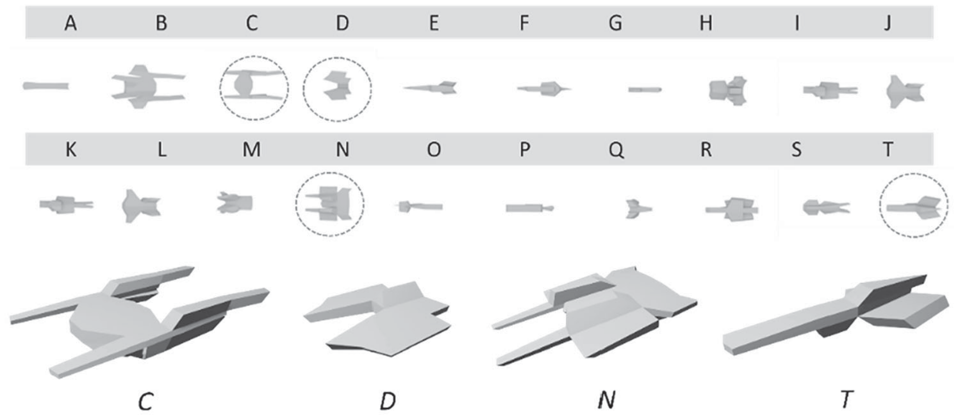

Figure 6. Correlating randomly generated shape 2D/3D samples.

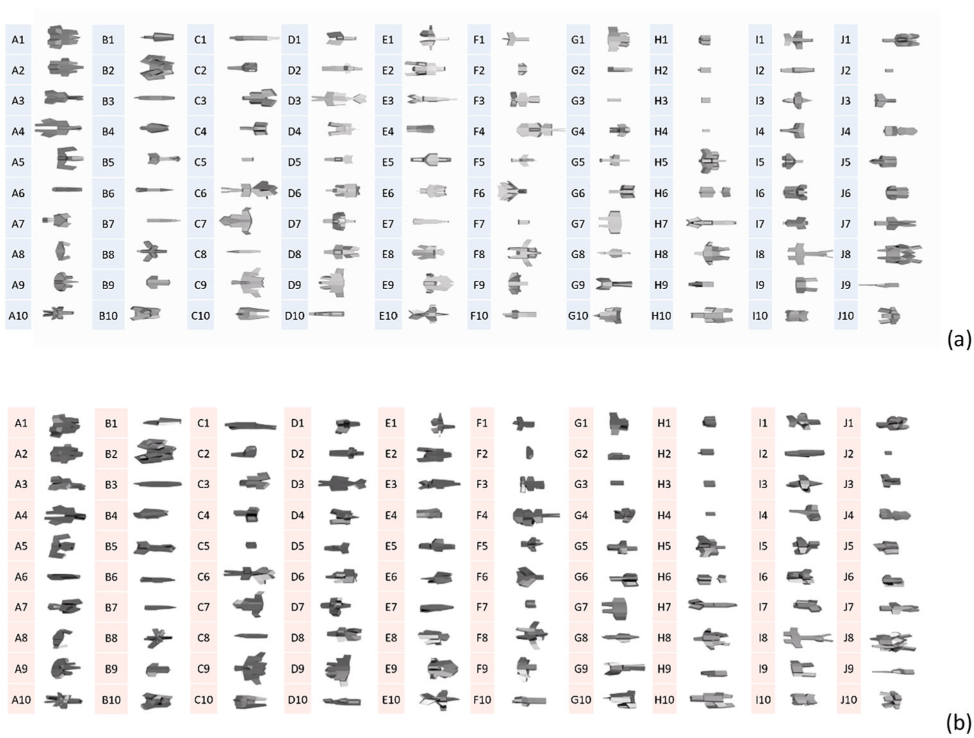

Figure 7. Numbering of (a) 2D and (b) 3D images (100 images shown as examples).

are less suitable for final mouse product designs. Larger surface areas are more compatible with hand ergonomics, making them more appropriate for mouse design. As shown in Figure 8, the projected areas of samples 1 through 200 were ranked in order. Samples toward the right of the figure have larger areas. The largest area was found in sample #111, followed by samples #23, #106, and #104.

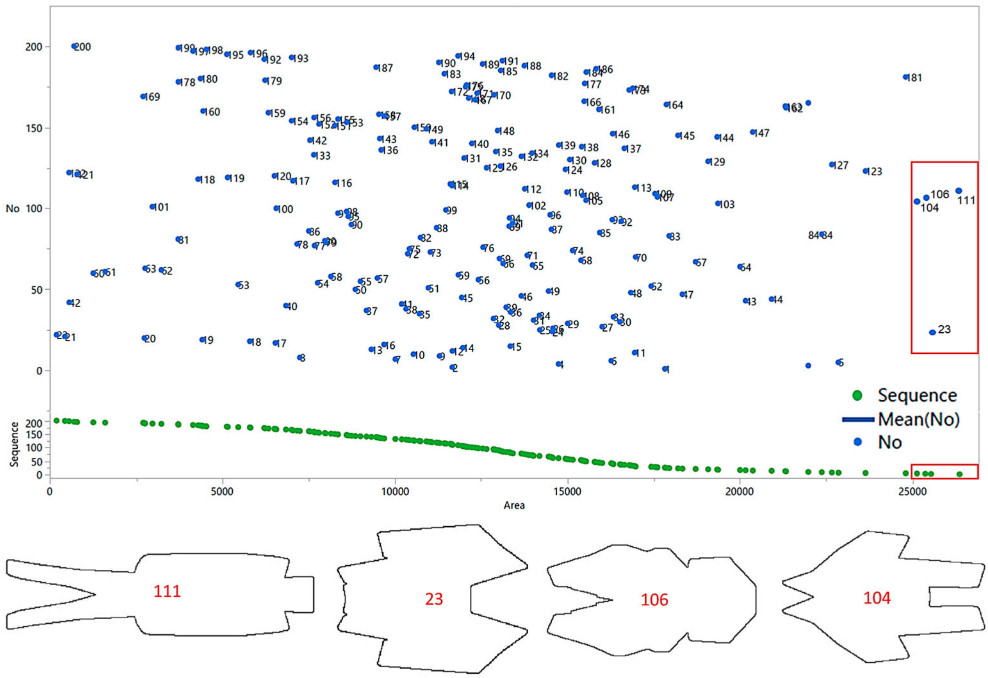

Figure 8. Measuring and ranking sample areas; top four: #111, #23, #106, #104.

4.3. Analyzing 2D samples using t-SNE

Next, the study applied the t-SNE machine learning technique to analyze the set of 200 generated samples. Simpler shapes are located toward the bottom right of the figure, while more complex geometric shapes are found toward the top left. For instance, sample #23, shown in Figure 8, corresponds to a complex cluster located in the left section of Figure 9, as indicated by the arrow in Figure 9.

The study discovered that 2D shapes that are too simple or elongated are less suitable for computer mouse designs, as such shapes are rare in the current market. The t-SNE method helps identify more suitable mouse shapes. Notably, though elongated mouse designs are rare, Genius released the Pen Mouse in 2011, which had such a shape. Therefore, t-SNE not only identifies common patterns but also inspires innovative design, enabling designers to break free from conventional thinking and discover novel design possibilities.

4.4. Converting generated samples from 2D/3D to surfaces

The study converts 2D/3D data to surfaces. The stage is a common step in product development, where conceptual shapes are transformed into tangible products. As shown in Figure 10, sample #23 (from dataset 35) can be traced from its 2D image to its corresponding 3D model from previously established correlations. Then we used Moi 3D software to convert the 3D model into a smooth surface, completing the top shell of the mouse.

Next, we design the bottom shell by connecting it to the upper shell using a flat 3D shape and a rectangular connector. This completed the external form of the 3D-generated mouse.

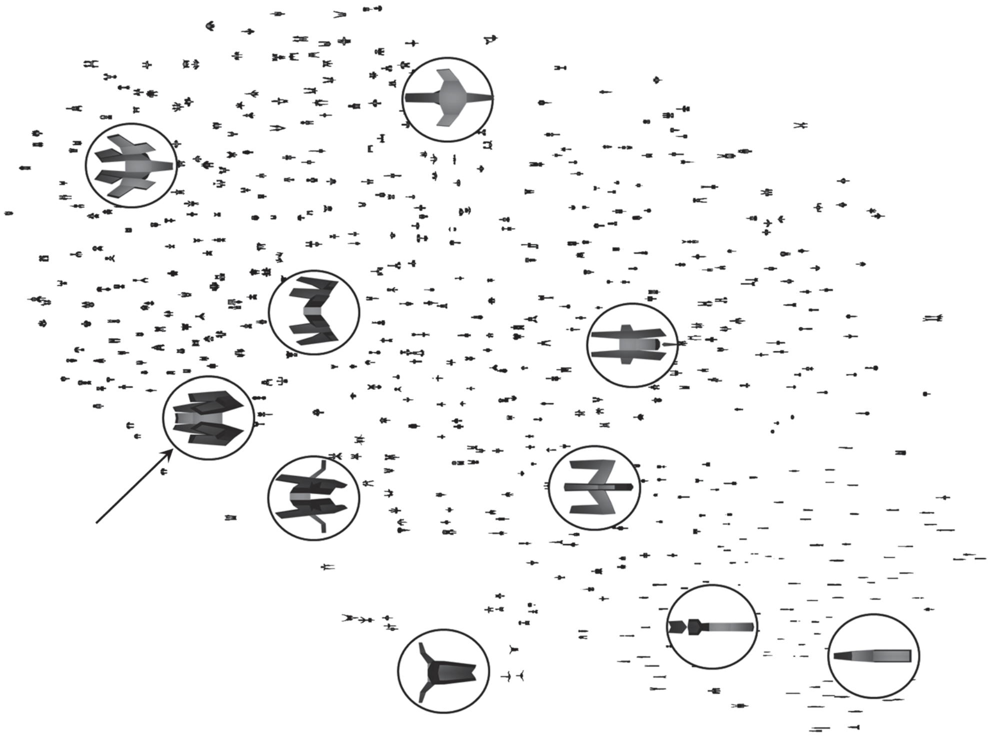

Figure 9. t-SNE analysis of 2D samples from randomly generated shapes.

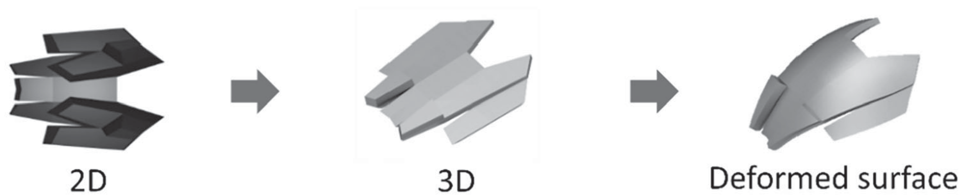

Figure 10. Converting the sample 2D into a surface (sample #23 from dataset 35).

However, no mechanical design or internal components have been included yet, so the mouse is only complete visually and not functionally operational at this stage. See Figure 11.

4.5. Brand style classification system for mouse designs

The study adopted the method from 'A data-driven approach to predicting and interpreting brand styles' (Wang and Chen 2025), which applied deep learning to assess brand styles. Two mouse brands are selected for the dataset: Genius and Mad Catz. Mad Catz's design features include wedge shapes, mechanical screw-like details, stacked geometries, transformer-inspired shapes, side wings, and distinctive scroll wheels. In contrast, Genius mice emphasise organic shapes, LED lighting, touch scroll wheels, mini mouse formats, unique materials, multiple buttons, and simple colour schemes. In this study, the generated

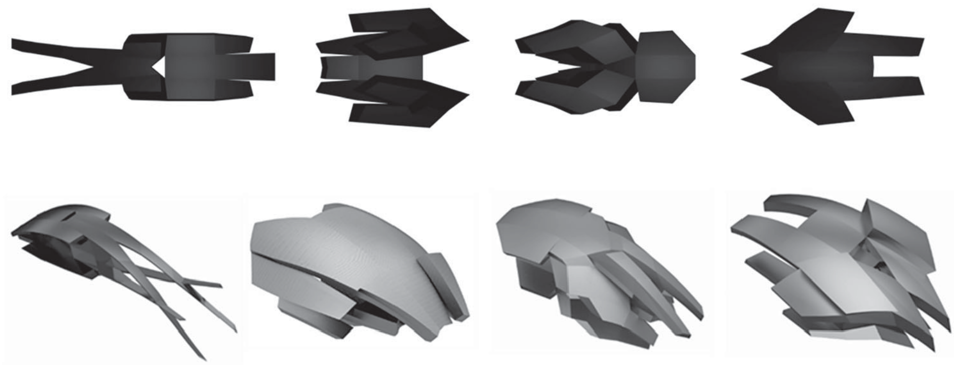

Figure 11. The sample' 2D to 3D surface conversion indicates a 3D mouse appearance.

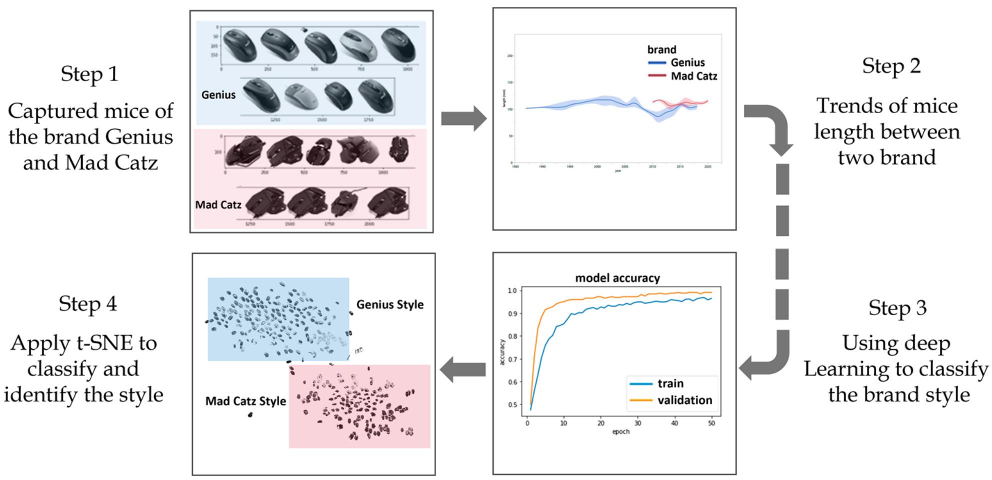

Figure 12. Brand style classification applied to generated mouse samples.

mouse designs aligned more closely with the Mad Catz style, characterised by features such as wedge shapes, transformer-like structures, and side wings. The classification procedure is illustrated in Figure 12.

In Wang and Chen's (2025) study, three student-designed mice all aligned with the Genius brand style. However, these randomly generated shape samples in this study predominantly exhibited characteristics of the Mad Catz brand. This is likely because the generative process favoured stacked, angular shapes, whereas human designers often prefer smooth, unified curves resembling the Genius style. Next, we use deep learning to predict brand style based on randomly generated designs.

4.6. Deep learning assessment and heatmap prediction

The study used deep learning-based heatmap analysis (Grad-CAM) to determine which brand style – Genius or Mad Catz – each generated sample most closely resembles. The four

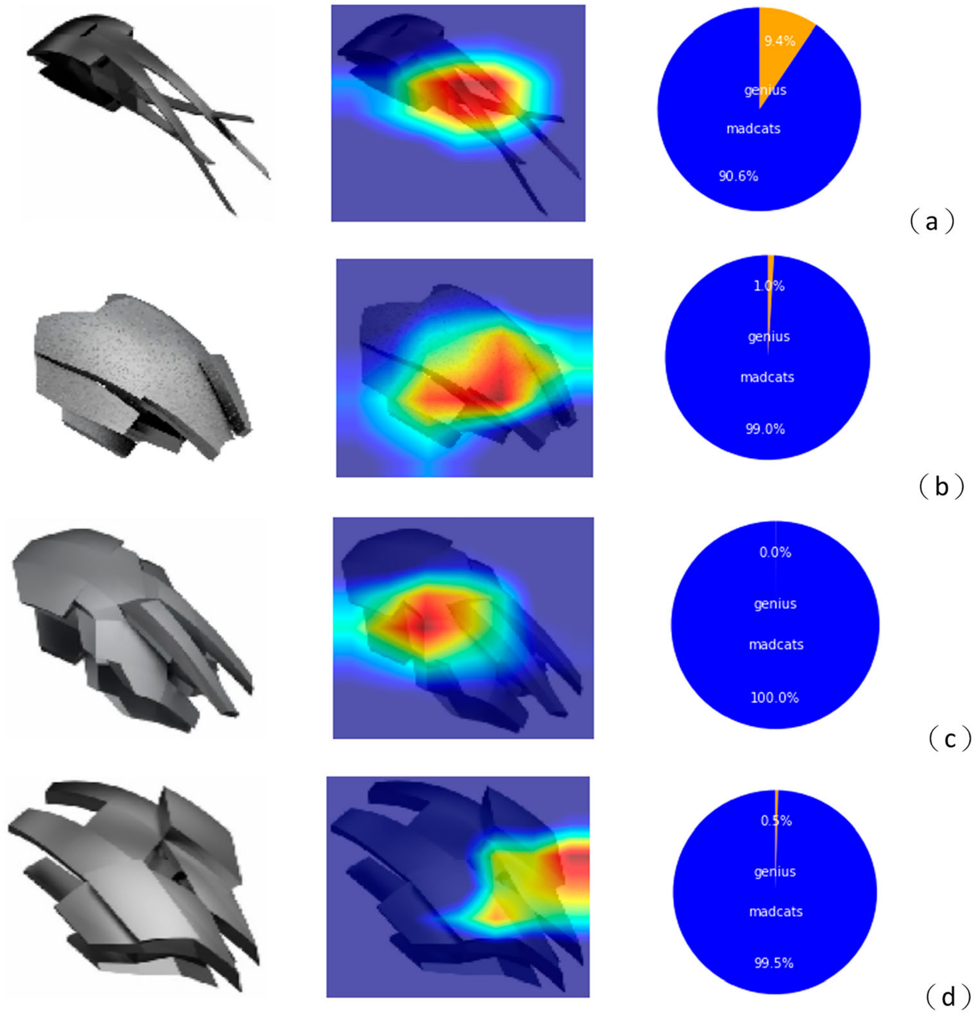

Figure 13. Apply Deep learning and Grad-CAM to predict the samples' brand style (a) Sample #111, (b) Sample #23, (c) Sample #106, (d) Sample #104.

samples tested were #111, #23, #106, and #104. Results are shown in Figure 13. Predictions for each sample are as follows:

A. Sample #111

Generated via shape randomisation, this sample has a 90.6% probability of being Mad Catz style and a 9.4% probability of being Genius. Features include wedge shapes, transformer-like shapes, and squid-like tentacles – matching the Mad Catz design traits.

B. Sample #23

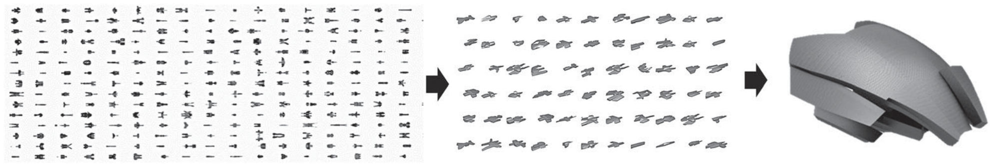

Figure 14. Integrating deep learning into shape random generation for optimised design selection based on brand style (Example: Sample #23 from Dataset 35).

Predicted to be 99% Mad Catz and 1% Genius. It features wedge shapes, transformer-inspired structures, stacked elements, and side wings.

C. Sample #106

Predicted to be 100% Mad Catz and 0% Genius. It also includes wedge shapes, transformer aesthetics, stacked forms, and squid-like tentacles.

D. Sample #104

Predicted to be 99.5% Mad Catz and 0.5% Genius. It features wedge shapes, transformer shapes, stacked geometry, and side wings.

On average, the generated samples have a 97.275% probability of belonging to the Mad Catz style and only 2.725% for the Genius style. In short, as long as the generation conditions include characteristics specific to a brand, deep learning can accurately classify the sample accordingly. Moreover, using Grad-CAM heatmaps, we can visualise why a sample was classified as Mad Catz. The highlighted areas correspond to design traits like wedge shapes, stacked structures, and transformer-like shapes. This heatmap information can serve as a valuable reference for designers creating products aligned with the Mad Catz brand style.

5. Discussion and analysis

5.1. Sample selection combining random shape generation and style classification system

This study begins by utilising shape random generation technology to create samples, assign identification numbers, and group them accordingly. Suitable samples are then selected through area measurement, followed by deep learning methods to evaluate the brand style of the new mouse designs. Finally, the machine learning technique t-SNE is applied to analyze the results.

The first half of the process involves generating random shapes, while the latter half utilises a style-specific classification system to replace human subjectivity in the selection of samples. Unlike the typically opaque decision-making process of human designers, the proposed style classification system makes the decision process transparent and clearly observable, as shown in Figure 14.

Through correlation and regression analysis, this study confirms that filtering by 2D plane area is beneficial in determining the final 3D surface. The correlation between the

2D plane and 3D surface assists in identifying optimised designs among a large set of generated samples. Additionally, the application of CAD helps convert 3D representations into surfaces, thereby completing the appearance of a computer mouse, which enables further evaluation through deep learning.

5.2. Predicting generated 3D sample's style using t-SNE

The method t-distributed Stochastic Neighbour Embedding (t-SNE) is a data visualisation technique. In machine learning, t-SNE is an unsupervised and nonlinear method used for exploring and visualising high-dimensional data (Van der Maaten and Hinton 2008). Furthermore, evaluations of new product designs using t-SNE are consistent with expert judgment, indicating that t-SNE can emulate human evaluation capabilities.

Similar to how Li, Xie, and Sha (2023) emphasse design representation for evaluating 3D generative outcomes, our method applies t-SNE to examine the stylistic performance of randomly generated product shapes. The purpose of this design experiment is to assess whether randomly generated shape mouse designs align with a specific brand's design DNA. The t-SNE method helps evaluate brand consistency in new mouse designs. The t-SNE module in Python is trained on datasets from Genius and Mad Catz, and used to classify whether the newly generated images resemble Genius or Mad Catz designs. As a result, Sample #23 is clustered within the Mad Catz brand style group in Figure 15, exhibiting Mad Catz characteristics such as wedge-shaped design, transformer-like form, stacked structures, and side wings.

Compared with previous studies, a notable difference emerges: randomly generated designs tend to be classified as Mad Catz, while student-designed mice are generally categorised under Genius. This result may be because randomly generated shapes often include stacked and wedge forms, whereas student designs favour singular, organic shapes. Grad-CAM heatmaps also help identify key brand style features. Therefore, this approach offers a rational and structured method for describing design styles that are often perceived as abstract and difficult to define.

5.3. Shape random generation and selection of distinct brand styles

This study proposes a new design methodology that integrates machine learning with shape random generation to develop products with specific brand styles. It provides an alternative to traditional design processes and can be used to verify the brand identity of randomly generated shape designs.

Due to resource and time constraints, this study focuses on two brands: Genius and Mad Catz, both of which have distinct styles. The t-SNE analysis shows an apparent clustering of samples into two groups based on these brand styles.

From an artistic perspective, individual artists typically maintain a consistent style. It's rare for an artist to switch from Realism to Fauvism suddenly. However, in commercial design, designers may adapt their style to align with a company's brand identity, which is shaped by factors such as usability, functionality, manufacturability, and ease of cleaning. Consequently, personal design styles are often overlooked, and many designers lack a distinct individual style. In some cases, it's not the designer but other departments in the company that influence brand style.

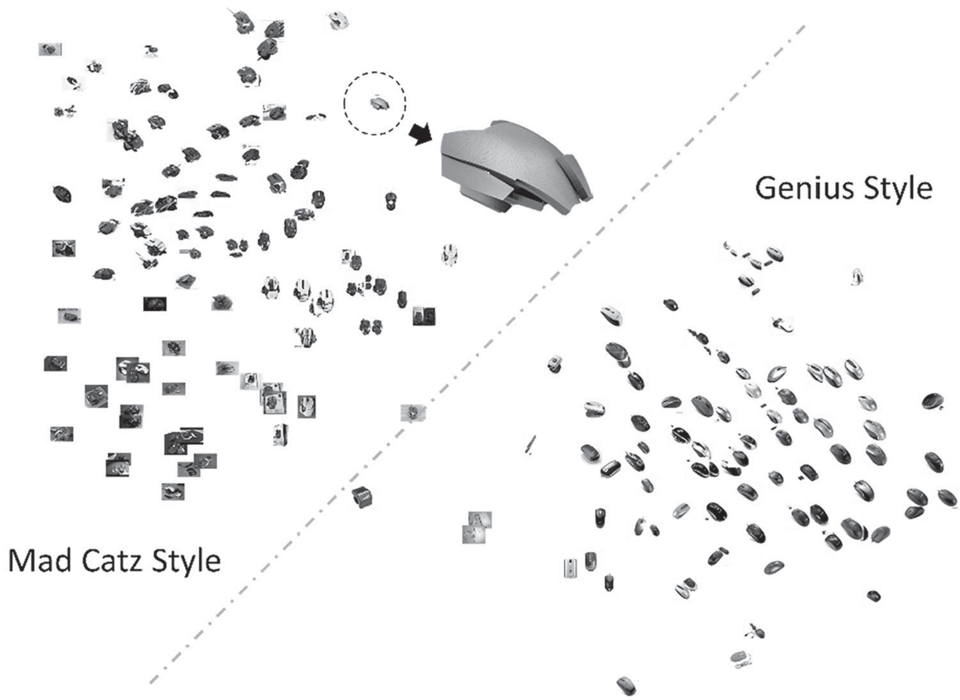

Figure 15. Using t-SNE to predict the brand style of selected samples (Example: Sample #23 from Dataset 35).

Furthermore, many companies adopt similar design thinking, resulting in homogeneous products that lack differentiation or a unique brand style – especially in industries with low design and manufacturing barriers. Similar concerns have been addressed in architecture, where AI-supported space production has been leveraged to reinforce local brand identity (Ekici and Çelik 2025). With the proposed tool, however, designers working in companies can generate a vast number of shapes and select those that match their unique brand style.

5.4. Applying AI to product design

This study explores the application of AI in product design. Key points are summarised below:

A. A random generation and filtering approach produces a large number of samples, filters out unsuitable ones, and retains those with the desired style.
B. The design process becomes transparent and standardised, rather than a black-box approach, clearly demonstrating how final samples are selected.
C. The process embodies computational thinking, which is a new way of thinking that can compare to human reasoning. It may even simulate a designer's creative logic – among thousands of design paths, one might mirror the reasoning of a particular designer.

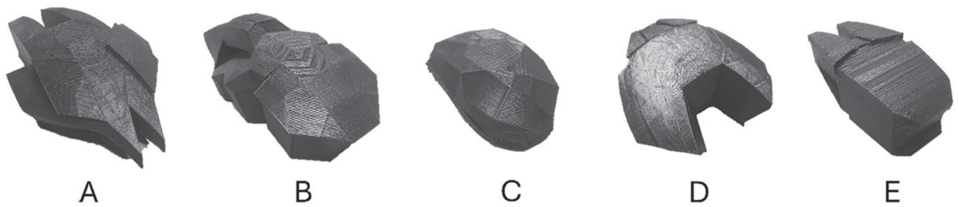

Figure 16. Five mice used for the survey of brand mice, comparing computer-generated and designer-created.

D. The generative brand style workflow includes four subsystems: generation, filtering, design, and classification. The input is a set of parameters, and the output is 2D/3D shapes. The filtering system ranks the 2D samples by area and selects the optimal candidates. The design system converts these into mouse prototypes, and the classification system evaluates the brand style based on the 3D images.
E. Unlike black-box generative AI tools (e.g. ChatGPT, Leonardo.ai, DALL·E, Stable Diffusion), this study presents a white-box AI application that is observable, understandable, and systematized. Recent work by Chen and Ruan (2025) similarly demonstrates a transparent generative – evaluative loop, where Stable Diffusion combined with LoRA supports analogical product design exploration and assessment.
F. Shape random generation for brand-style products can go beyond computer peripherals and be applied to automotive design, furniture, fashion, home appliances, etc., enabling companies to explore diverse shapes and select optimised brand style consistent designs. It can also help designers break habitual thinking and unlock novel design possibilities through machine learning analysis.

5.5. Survey on randomly shape generated vs. designer-created brand mouse designs

This study concludes with a user survey comparing mouse designs randomly generated and those designed by human designers, with a focus on visual appearance and usability. The goal is to determine whether the use of random shape generation for input device development can meet market needs. Participants were frequent computer mouse users, excluding individuals with severe upper limb disabilities or neuromuscular conditions. The survey was conducted with 31 participants, primarily second-year students, during a product design course.

The study, which utilised the concept of shape random generation and selection, identified proper shapes from the database. An anonymous survey compared randomly generated and designer-created mice using descriptive statistics and two-sample t-tests, performed with Minitab software. The study includes five mice in this survey: A to E. Mouse A was used only for survey testing and excluded from statistical analysis. Mice B and D were based on computer random generation; mice C and E were designer-created (Figure 16).

A two-sample t-test was used to compare ratings for appearance and usability. Results showed that the randomly computer-generated group received significantly higher appearance scores than the designer-created group (3.21 vs. 2.68), with a T-value of 2.54

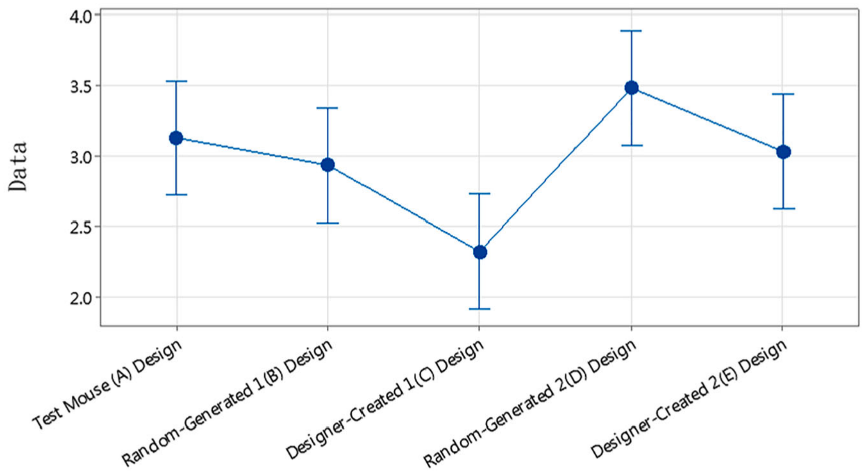

Figure 17. Appearance score differences between randomly computer-generated and designer-created mice via a two-sample t-test.

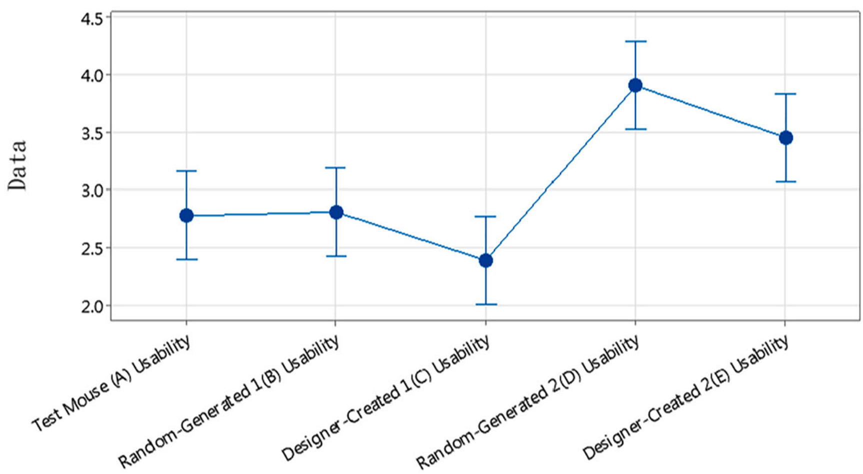

Figure 18. Usability score differences between randomly computer-generated and designer-created mice via a two-sample t-test.

and a p-value of 0.012. Since $p < 0.05$, the null hypothesis was rejected, confirming a statistically significant difference in visual appeal (Figure 17).

The same analysis was applied to usability. The randomly computer-generated group received higher average scores (3.35 vs. 2.92), with a T-value of 2.06 and p-value of 0.041, also indicating statistical significance (Figure 18).

In summary, the study found that randomly computer-generated mice outperformed designer-created ones in both appearance and usability, with statistically significant differences. These findings challenge the traditional belief that human design always surpasses computer-generated shape, demonstrating that random shape generation can yield design quality comparable to human efforts in input device development.

6. Conclusion

This study proposed a new design approach that integrates random shape generation, brand style classification, and machine learning evaluation to support industrial product design. The method responds directly to the four challenges outlined in the introduction. First, it addresses the difficulty of converting 2D images into 3D models by employing systematic random shape generation and CAD-based surface modelling. Second, it supports the development of innovative concepts by generating a wide variety of design alternatives in the early ideation phase. Third, it introduces AI-based filtering and selection methods to evaluate large sample sets and identify optimal designs. Fourth, it verifies whether new products embody specific brand styles through a deep learning classification system and heatmap visualisation. Through empirical investigation and surveys, the study demonstrates that machine learning can effectively assist designers in producing brand-style-aligned outputs and that randomly generated designs can surpass human-created ones in both appearance and usability. The findings confirm the potential of this method for industrial applications, particularly in input device design, while also suggesting broader opportunities in fields such as automotive, furniture, and home appliances. Beyond these immediate contributions, the approach provides a foundation for expanding into design decision-making optimisation, human factors evaluation, and the development of systematic design knowledge frameworks. A principal contribution of this study is a novel, general, and auditable verification protocol that pairs generation by generative AI with evaluation by deep-learning classification and saliency maps, thereby transforming emotional design from a subjective domain into measurable design criteria, scaling verification to large candidate sets, and enabling quantitative benchmarking of generative systems against consumer-recognition goals. Ultimately, it contributes not only to practical product development but also to advancing the integration of artificial intelligence into brand-style product design.

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Disclosure statement

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Ethical approval statement

Ethical approval was obtained from the Human Research Ethics Review Committee at the National Chengchi University (Submission number: NCCU-REC-202506-E108).

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